

Resolution Geometry

Quantum Metric Selection: Task Locks, Nonparametric Identification and Wall Specialization of Petz Metrics

Matěj Rada

MatejRada@email.cz

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Abstract

In finite-dimensional quantum experiments, the metric module of Resolution Geometry is not unique. Monotonicity under completely positive trace-preserving (CPTP) maps admits the whole Petz family of monotone metrics, and on commuting tasks every member reduces to the classical Fisher metric. This paper closes the quantum metric layer. The member is selected by the declared operational task:

- local estimation with optimal measurement selects SLD (Braunstein–Caves);
- asymmetric (Stein) testing selects BKM, the Hessian of the Umegaki relative entropy;
- symmetric (Chernoff) testing selects Wigner–Yanase.

The Petz function is also identified from data without any candidate panel: commuting tasks fix the scale, and off-diagonal qubit tasks give $f(t)$ pointwise. A finite-panel fit, as in the earlier battery, names a winner even on data from an off-panel metric, and it is resolution-relative: admixtures below $w_{\text{res}} = \text{noise/gap}$ are invisible.

At rank (support) walls, a Petz metric specializes on tangents coupling support and kernel iff $f(0) > 0$. SLD tends to the Fubini–Study metric (times 4) and WY to 4. BKM diverges like $\log(1/\varepsilon)$, Heinz(1/4) like $\varepsilon^{-1/4}$, and RLD like ε^{-1} . Metric selection selects no transport.

A ten-check battery passes 10/10 on five seeds and three mutation canaries are caught.

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1 The quantum observation system

Let $\theta \mapsto \rho_\theta$ be a smooth family of full-rank density matrices on $\mathbb{C}^d$, observed through a declared family of measurements, so that $F(\theta)$ is the joint law of their outcomes. When the family is tomographically complete, the kernel groupoid is equality of states, the stabilizer contains the unitary covariance of the protocol, and natural modules are pullbacks of structures on the state space that are invariant under the declared relabelings. Quantum Markov invariance is monotonicity under CPTP maps.

Definition 1.1 (Petz metric). For an operator monotone $f : (0, \infty) \rightarrow (0, \infty)$ with $f(t) = tf(1/t)$ and $f(1) = 1$, and a traceless Hermitian tangent A written in the eigenbasis $\rho = \sum_i \lambda_i |i\rangle\langle i|$,

$$g_f(\rho; A, A) = \sum_{i,j} \frac{|A_{ij}|^2}{\lambda_j f(\lambda_i/\lambda_j)}.$$

The panel used throughout is

$$f_{\text{SLD}} = \frac{1+t}{2}, \quad f_{\text{BKM}} = \frac{t-1}{\log t}, \quad f_{\text{WY}} = \left(\frac{1+\sqrt{t}}{2}\right)^2, \quad f_{\text{RLD}} = \frac{2t}{1+t}, \quad f_{\text{Heinz}} = \frac{t^{1/4} + t^{3/4}}{2}.$$

Imported Theorem 1.2 (Petz classification; [10, 6]). *A Riemannian metric on full-rank states is monotone under every CPTP map iff it is cg_f for such an f and some $c > 0$. On commuting states and tangents all g_f coincide with the classical Fisher metric.*

Proposition 1.3 (Family gate; QS01, QS02). *Monotonicity is a gate, not a selector.*

- On 200 commuting tasks the panel spread is 0.
- Under 400 random three-Kraus channels no panel member increased (maximal change < 0).
- The symmetric, normalized, non-operator-monotone $f(t) = (t^2 + t^{-1})/2$ increased by 0.87.

2 Task selection

Theorem 2.1 (Task lock). *For a declared local task at ρ in direction A , the task fixes the Petz member:*

1. Estimation with optimal measurement: $\max_{\text{POVM}} F_{\text{cl}} = g_{\text{SLD}}(\rho; A, A)$, attained by projective measurement in the SLD eigenbasis [7, 3].
2. Asymmetric testing (quantum Stein lemma): the optimal type-II exponent is $D(\rho \parallel \sigma)$ [8, 9], and $D(\rho + hA \parallel \rho) = \frac{h^2}{2} g_{\text{BKM}}(\rho; A, A) + O(h^3)$.
3. Symmetric testing (quantum Chernoff bound): the exponent is $\xi = -\log \min_{0 \leq s \leq 1} \text{Tr } \rho^s \sigma^{1-s}$ [2], and $\xi(\rho, \rho + hA) = \frac{h^2}{8} g_{\text{WY}}(\rho; A, A) + O(h^3)$, with optimum $s = 1/2$ [5].

Since g_{SLD} is the minimal Petz metric, every other member strictly exceeds the measured Fisher information on noncommuting tangents.

Proof. (1) Braunstein–Caves bound and its attainment. (2) Differentiate D twice: the second derivative of $\text{Tr } \rho \log \rho$ in direction A is the BKM form. (3) Expand $\text{Tr } \rho^{1/2} \sigma^{1/2}$ to second order; the Wigner–Yanase skew term appears, and symmetry of the local problem gives $s = 1/2$. $\square$

check	task	selected	deviation from selected	gap to others
QS03	optimal estimation	SLD	$< 10^{-12}$	$\geq 7 \times 10^{-4}$
QS04	Stein testing	BKM	5.3×10^{-8}	7.6% (to SLD)
QS05	Chernoff testing	WY	3.4×10^{-4} ($h = 10^{-4}$)	2.4%

The fixed- s Chernoff exponent does not recover WY (canary).

Corollary 2.2. *The quantum metric module of RG is task-relative. A protocol that declares its task has a unique metric up to scale. A protocol that does not declare one has the Petz family, and certificates abstain on the member.*

3 Identification from data

Theorem 3.1 (Nonparametric identification). *Let responses $y = \kappa g_f(\rho; A, A)$ be measured with unknown scale κ and unknown Petz function f .*

1. *Commuting tasks determine κ , since all members agree there.*
2. *For $\rho = \text{diag}(\lambda_1, \lambda_2)$ and $A = \sigma_x/\sqrt{2}$,*

$$f(t) = \frac{2\kappa|A_{12}|^2}{\lambda_2 y}, \quad t = \lambda_1/\lambda_2.$$

3. *Responses at a set T of ratios determine f on T , and on $\overline{T}$ by continuity. Every operator monotone f is continuous, so dense T determines f .*

Proof. For diagonal ρ and off-diagonal A , $g_f = |A_{12}|^2 \left(\frac{1}{\lambda_2 f(t)} + \frac{1}{\lambda_1 f(1/t)} \right)$. Since $f(1/t) = f(t)/t$, the second term equals the first, and solving gives the formula. $\square$

No-Go 3.2 (Panel locks; QS06). Generate responses from the off-panel Petz function $f^* = 0.6 f_{\text{BKM}} + 0.4 f_{\text{WY}}$, with relative noise 10^{-5} . Then:

- Theorem 3.1 recovers the scale to 7×10^{-7} and f^* to 2.1×10^{-5} ;
- every panel member is rejected;
- a panel fit still names BKM.

A finite panel is a model-selection device, not a lock. With noise of 3×10^{-2} the reconstruction fails its own gate (canary).

Proposition 3.3 (Resolution band; QS07). *Let the noise level be ν and let Δ be the median relative response gap between two members. Then a mixture $(1-w)f_1 + wf_2$ is indistinguishable from f_1 iff $w \lesssim w_{\text{res}} = \nu/\Delta$. For BKM data with 40 tasks at $\nu = 2 \times 10^{-5}$: $w_{\text{res}} = 2.3 \times 10^{-4}$. An admixture of $0.1 w_{\text{res}}$ fits as well as BKM (loss 3.4×10^{-10}); one of $10 w_{\text{res}}$ is rejected (loss 8.1×10^{-8}). A metric certificate states a band of f .*

4 Walls

Theorem 4.1 (Wall specialization). *Let $\rho_\varepsilon \rightarrow \rho_0$ approach a support wall, and let A couple the support with the kernel. Then*

$$g_f(\rho_\varepsilon; A, A) \rightarrow \sum_{i \in \text{supp}, j \in \text{ker}} \frac{2|A_{ij}|^2}{\lambda_i f(0)} \quad \text{iff } f(0) > 0,$$

and diverges otherwise. Writing $c(\lambda, \varepsilon) = 1/(\varepsilon f(\lambda/\varepsilon))$:

- *if $f(0) > 0$, then $f(t) \sim t f(0)$ as $t \rightarrow \infty$, so $c \rightarrow 1/(\lambda f(0))$;*
- *for BKM, $c \sim \log(1/\varepsilon)$;*
- *for Heinz(1/4), $c \sim \varepsilon^{-1/4}$;*

- for RLD, $c \sim \varepsilon^{-1}$.

SLD specializes to the pure-state wall as $4\times$ Fubini–Study [4].

Proof. $f(\lambda/\varepsilon) = (\lambda/\varepsilon)f(\varepsilon/\lambda)$ and $f(\varepsilon/\lambda) \rightarrow f(0)$, so $c = 1/(\lambda f(\varepsilon/\lambda)) \rightarrow 1/(\lambda f(0))$; the rates follow from the expansions of each f at 0. For pure states, SLD is the minimal metric, whose limit is $4\times$ Fubini–Study. $\square$

Measured on $\rho_\varepsilon = \text{diag}(1 - \varepsilon, \varepsilon)$ with $A = \sigma_x/\sqrt{2}$ (QS08, QS09):

- SLD $\rightarrow 2.000000$ and WY $\rightarrow 3.9992$ (theory 2 and 4);
- fitted powers: RLD 0.99999, Heinz 0.2511, BKM log-coefficient 0.9998;
- the pure-state SLD limit matches $4\times$ Fubini–Study to $< 10^{-12}$ on 100 random states.

Definition 4.2 (Re-discrimination at a wall). On the wall stratum, the metric module is selected again by the specialized task family. There are five outcomes:

- LOCKED: a finite task metric, identified within its band;
- DIVERGENT: $f(0) = 0$ on tangents coupling support and kernel. The task distinguishes perfectly across the wall; for example, the Stein exponent is infinite when supports differ;
- VACUOUS: no nonzero admissible tangent survives (for instance, trace-free tangents inside a one-dimensional support);
- UNDERDETERMINED: the band contains several members;
- INCONSISTENT: no Petz member fits.

Only LOCKED carries a metric across the wall. This is the quantum form of Hellinger specialization (L24.1): Bures/SLD distinguishability is continuous in trace norm, relative-entropy distinguishability is not.

No-Go 4.3 (No transport from the metric; QS10). For $\rho_\theta = U_\theta \rho U_\theta^\dagger$, two latent generators that differ by the stabilizer of ρ give the same tangent and the same value for every Petz member (difference 0), while the generators differ by 3.4. Metric selection leaves the transport undetermined; see the canonical transport theorem of [1].

5 Ladder insertion

Rung	Statement	Status	Ref.
Q1	CPTP monotonicity is a family gate (Petz); commuting tasks collapse to Fisher	imported	Thm. 1.2
Q2	Task lock: estimation $\rightarrow$ SLD, Stein $\rightarrow$ BKM, Chernoff $\rightarrow$ WY	imported / forced	Thm. 2.1
Q3	Nonparametric identification of f ; panel locks are not locks	forced (no-go)	Thm. 3.1
Q4	Metric certificates are resolution bands	forced	Prop. 3.3
Q5	Wall specialization iff $f(0) > 0$; re-discrimination outcomes	forced	Thm. 4.1
Q6	Metric selection does not select transport	no-go	QS10

6 Non-claims

- The theorems are stated for finite dimension. The battery tests qubits, which suffice for the pointwise identification of f .
- Synthetic responses establish the identification procedure, not which task a physical laboratory performs. The task is a declaration.
- No statement is made about infinite-dimensional or non-faithful states beyond the wall theorem.

A Reproducibility and audit

`rg_quantum_metric_attack.py` passes 10/10 on seeds 20261002, 1, 42, 777 and 2026. The canaries remove, in turn:

- the non-monotone control (QS02);
- the optimization over s (QS05);
- the noise budget (QS06).

Each targeted check fails.

Corrections before registration:

1. QS09 first removed the phase part of $\langle\psi|d\psi\rangle$ instead of the normalization part.
2. QS07 first used a fixed admixture $w = 10^{-3}$, which is $4w_{\text{res}}$ and therefore resolvable; the band is now computed from the data.

Audit of the earlier quantum battery:

- Its BKM lock fitted BKM-generated data with a five-member panel, so it could not fail (No-Go 3.2).
- Its “finite strata remain discriminating near wall” measured the divergence of the $f(0) = 0$ members, not discrimination (Theorem 4.1).
- Its vacuous-wall check was the literal **True**.

Its correct findings are retained here: Umegaki Hessian = BKM, monotonicity as a family gate, commuting collapse, and abstention for observational duplicates.

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